KubeCon CloudNativeCon Japan 2026

AIOps: (near) Zero-Touch Production Rollout Fixes

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- ★ Technical Lead, CNCF DevEx TAG
- ★ Agentic AI Foundation Ambassador
- ★ From Belgium 🇧🇪 / Live in Switzerland
- ★ 🗣️ English, Dutch, French, Italian

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CrowdStrike outage halts air, rail operations around Japan

THE ASAHI SHIMBUN

July 20, 2024 at 13:06 JST

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list



A long line forms at the Jetstar Airways counter at Fukuoka Airport on July 19. (Naoki Nakayama)

Parts of Japan were thrown into chaos by the CrowdStrike outage that triggered a series of abnormal shutdowns of Windows-based computers.

From 2h ago

08.47 CEST

Crowdstrike identifies issue behind worldwide outage



Josh Taylor

Cybersecurity software firm Crowdstrike has posted in its support updates that it has identified the issue behind today's massive, global Windows outage.

It is an issue with "content deployment" that has since had those changes reverted and has advised a workaround for affected users.

We are still waiting for confirmation or a statement from Crowdstrike's media team. It's almost midnight in the US, so it's not clear when we might get more detail.

+ Show more

What could have mitigated the impact?

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6. Template Instances should have staged deployment

Findings: Each Template Instance should be deployed in a staged rollout.

Mitigation: The Content Configuration System has been updated with additional deployment layers and acceptance checks

Staged deployment mitigates impact if a new Template Instance causes failures such as system crashes, false-positive detection volume spikes or performance issues. New Template Instances that have passed canary testing are to be successively promoted to wider deployment rings or rolled back if problems are detected. Each ring is designed to identify and mitigate potential issues before wider deployment. Promoting a Template Instance to the next successive ring is followed by additional bake-in time, where telemetry is gathered to determine the overall impact of the Template Instance on the endpoint.


Progressive Delivery
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Japan 2026
準備
試飲
問題発見
安心

GitOps



Git is the single source of truth





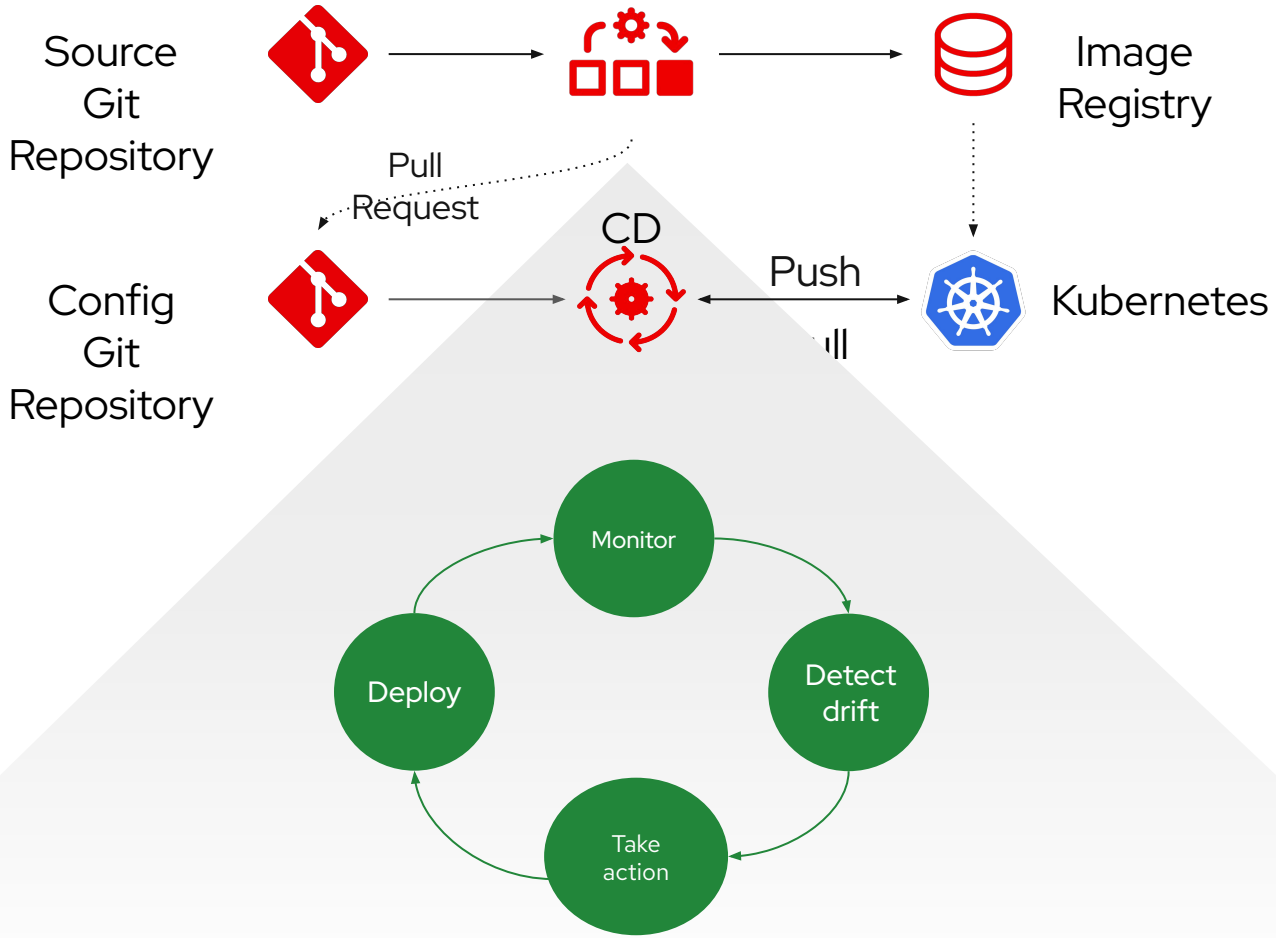
Treat everything as code





Operations through Git workflows

GitOps Application Delivery Model



By default:

a big bang / all or nothing release

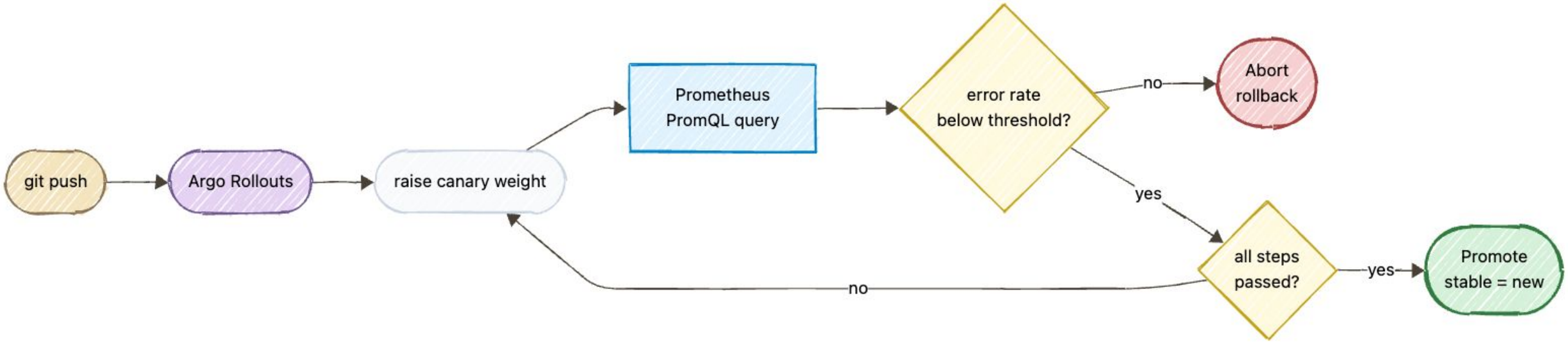


Argo Rollouts



flagger




git push
Argo Rollouts
raise canary weight
Prometheus
PromQL query
error rate
below threshold?
no
Abort
rollback
yes
all steps
passed?
yes
Promote
stable = new
no


HEY KID
WANNA SEE SOME YAML?
imgflip.com

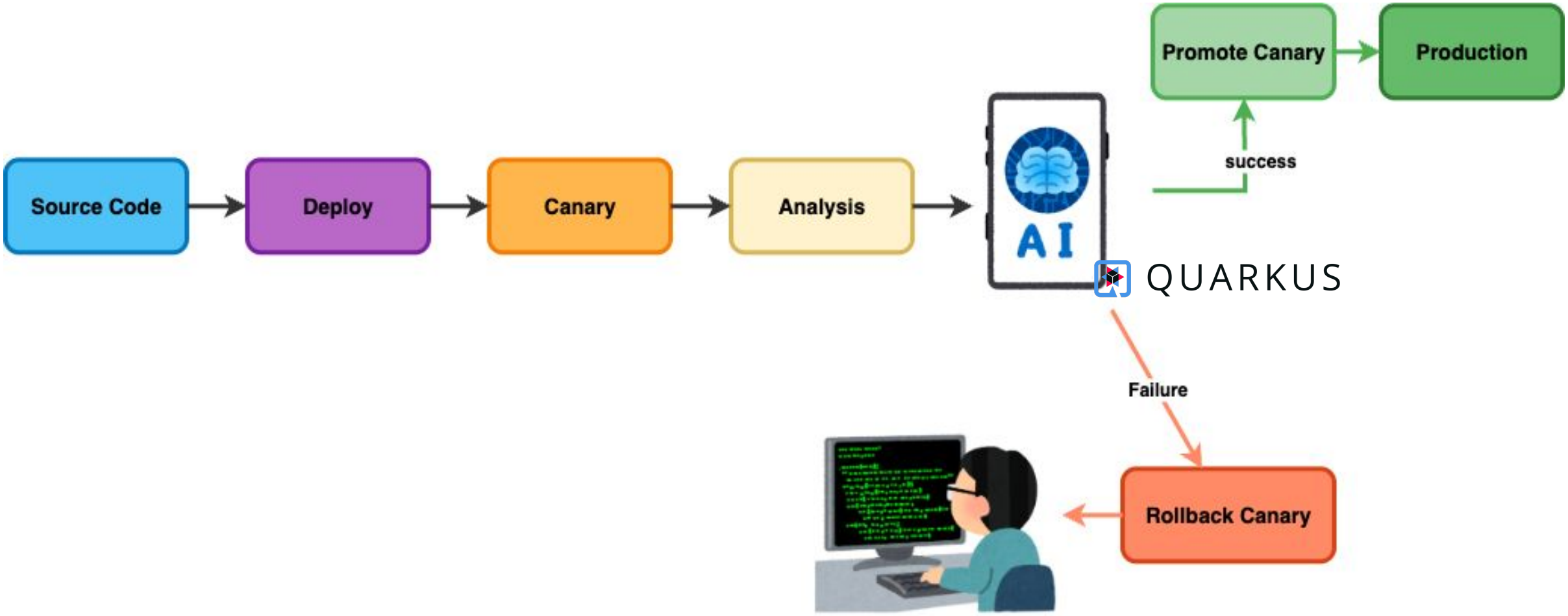
```yaml
apiVersion: argoproj.io/v1alpha1
kind: AnalysisTemplate
metadata:
name: success-rate
spec:
args:
- name: service-name
metrics:
- name: success-rate
  interval: 10s
  successCondition: len(result) == 0 || result[0] >= 0.95
  failureLimit: 2
  provider:
    prometheus:
      address:
https://internal:demo@prometheus.istio-system.svc.cluster
.local:9090
      query: |
        sum(irate(istio_requests_total{
          reporter="source",
          destination_service=~"{{args.service-name}}",
          response_code!~"5.*"}[30s])
        )
```

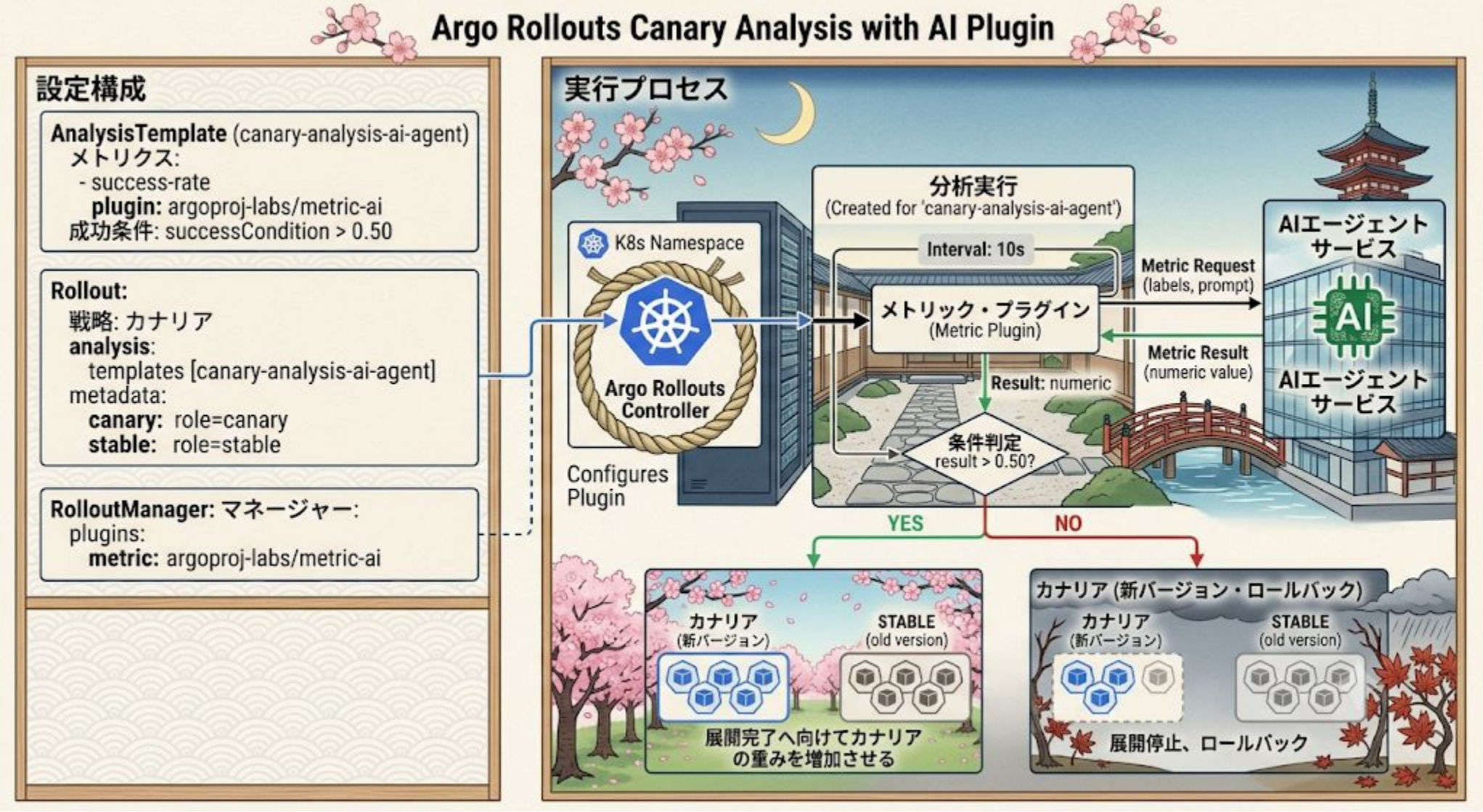
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    prometheus:
      address:
https://internal:demo@prometheus.istio-system.svc.cluster
.local:9090
      query: |
        sum(irate(istio_requests_total{
          reporter="source",
          destination_service=~"{{args.service-name}}",
          response_code!~"5.*"}[30s])
        )
```



Write custom PromQL

Use AI to analyze failures

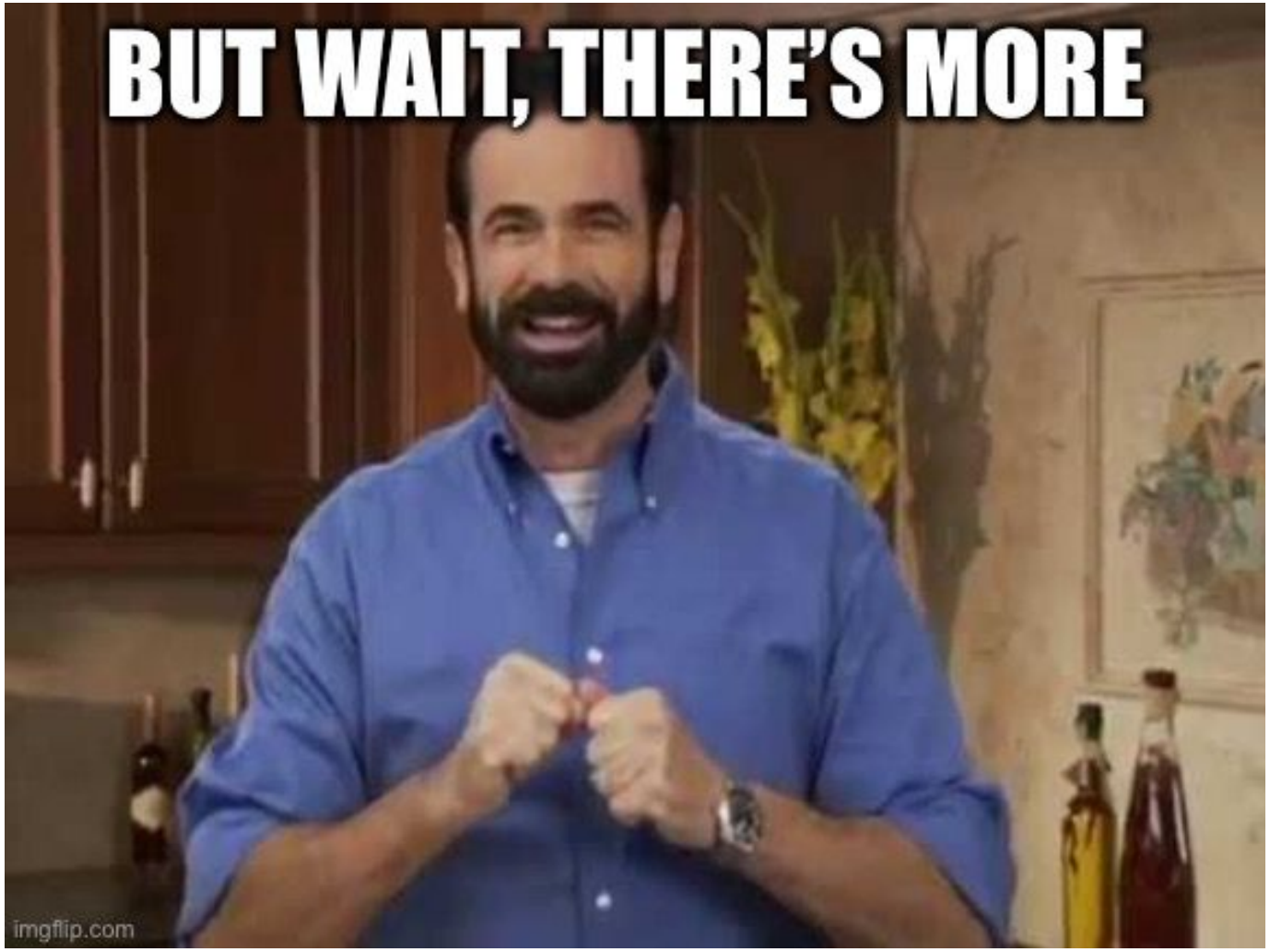

Source Code
Deploy
Canary
Analysis
AI
QUARKUS
Promote Canary
Production
success
Failure
Rollback Canary

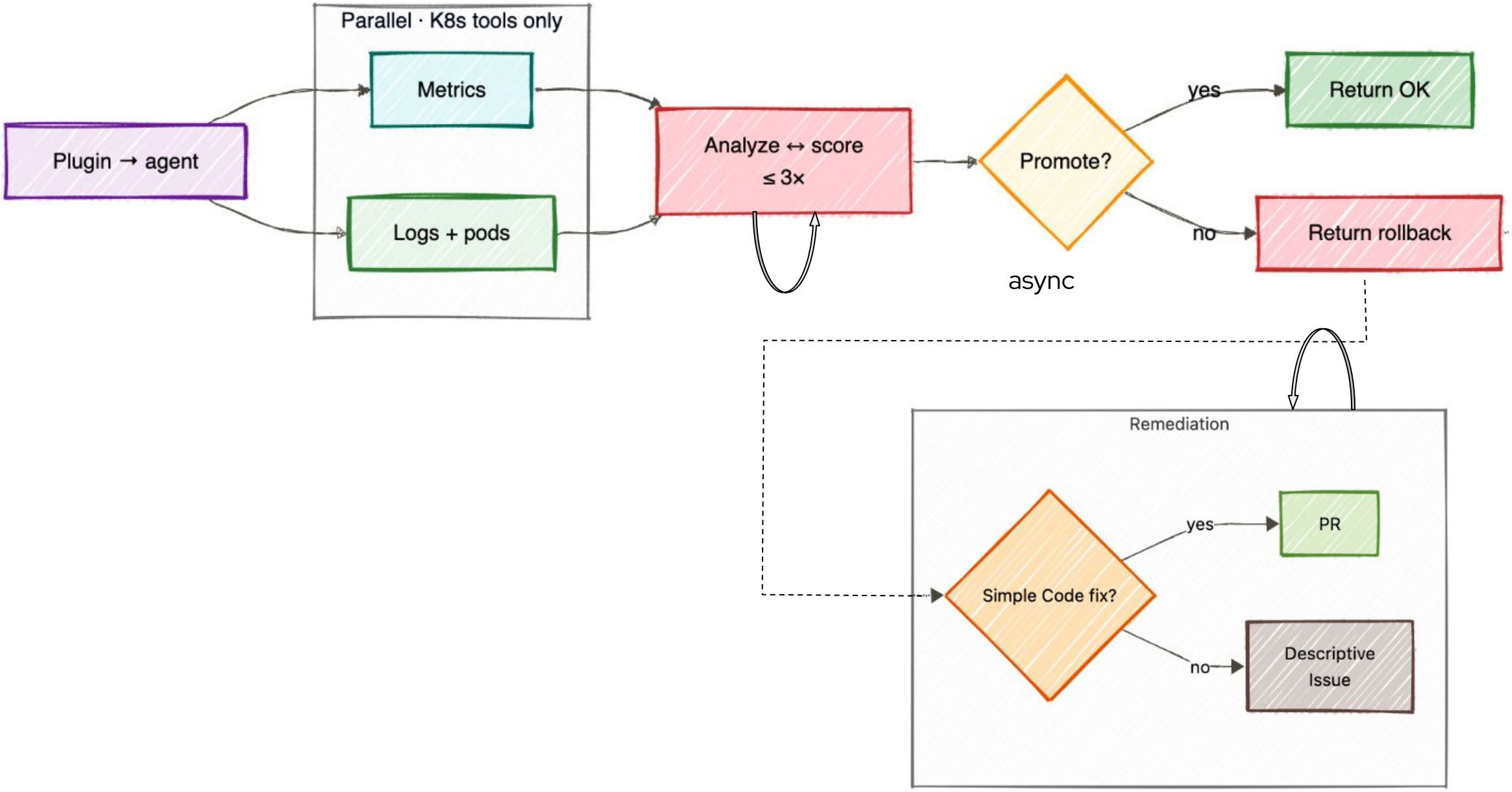

Argo Rollouts Canary Analysis with AI Plugin
設定構成
AnalysisTemplate (canary-analysis-ai-agent)
メトリクス:
- success-rate
plugin: argoproj-labs/metric-ai
成功条件: successCondition > 0.50
Rollout:
戦略: カナリア
analysis:
templates [canary-analysis-ai-agent]
metadata:
canary: role=canary
stable: role=stable
RolloutManager: マネージャー:
plugins:
metric: argoproj-labs/metric-ai
実行プロセス
K8s Namespace
Argo Rollouts Controller
Configures Plugin
分析実行
(Created for 'canary-analysis-ai-agent')
Interval: 10s
メトリック・プラグイン
(Metric Plugin)
Metric Request
(labels, prompt)
Metric Result
(numeric value)
Result: numeric
条件判定
result > 0.50?
AIエージェント
サービス
AI
AIエージェント
サービス
YES
NO
カナリア
(新バージョン)
STABLE
(old version)
展開完了へ向けてカナリア
の重みを増加させる
カナリア (新バージョン・ロールバック)
カナリア
(新バージョン)
STABLE
(old version)
展開停止、ロールバック

```
apiVersion: argoproj.io/v1alpha1
kind: RolloutManager
metadata:
  name: argo-rollouts
spec:
  plugins:
    metric:
      - name: argoproj-labs/metric-ai
        location:
https://github.com/argoproj-labs/rollouts-plugin-metric-ai/releases/
download/v0.0.1/rollouts-plugin-metric-ai-linux-amd64
```

```yaml
canary:
  analysis:
    startingStep: 1
    templates:
    - templateName: canary-analysis-ai-agent
  canaryMetadata:
    labels:
      role: canary
  stableMetadata:
    labels:
      role: stable
```

```yaml
apiVersion: argoproj.io/v1alpha1
kind: AnalysisTemplate
metadata:
 name: canary-analysis-ai-agent
spec:
  metrics:
  - interval: 10s
    name: success-rate
    provider:
      plugin:
        argoproj-labs/metric-ai:
          agentUrl: http://kubernetes-agent:8080
          stableLabel: role=stable
          canaryLabel: role=canary
          extraPrompt: ignore aesthetic changes
    successCondition: result > 0.50
```


BUT WAIT, THERE'S MORE
imgflip.com


Parallel · K8s tools only
Metrics
Logs + pods
Plugin → agent
Analyze ↔ score
≤ 3x
Promote?
yes
Return OK
no
Return rollback
async
Remediation
Simple Code fix?
yes
PR
no
Descriptive
Issue

```yaml
apiVersion: argoproj.io/v1alpha1
kind: AnalysisTemplate
metadata:
 name: canary-analysis-ai-agent
spec:
  metrics:
  - interval: 10s
    name: success-rate
    provider:
      plugin:
        argoproj-labs/metric-ai:
          agentUrl: http://kubernetes-agent:8080
          stableLabel: role=stable
          canaryLabel: role=canary
          extraPrompt: ignore aesthetic changes
          githubUrl: ...github.com/kdubois/argo-rollouts-quarkus-demo
```

Demo

Lessons learned:

Performance was an interesting challenge:

went from one AI service to agentic system: parallel & async agents

LLM choice + “context engineering” + tool calling

especially for PR creation

Complexity vs portability

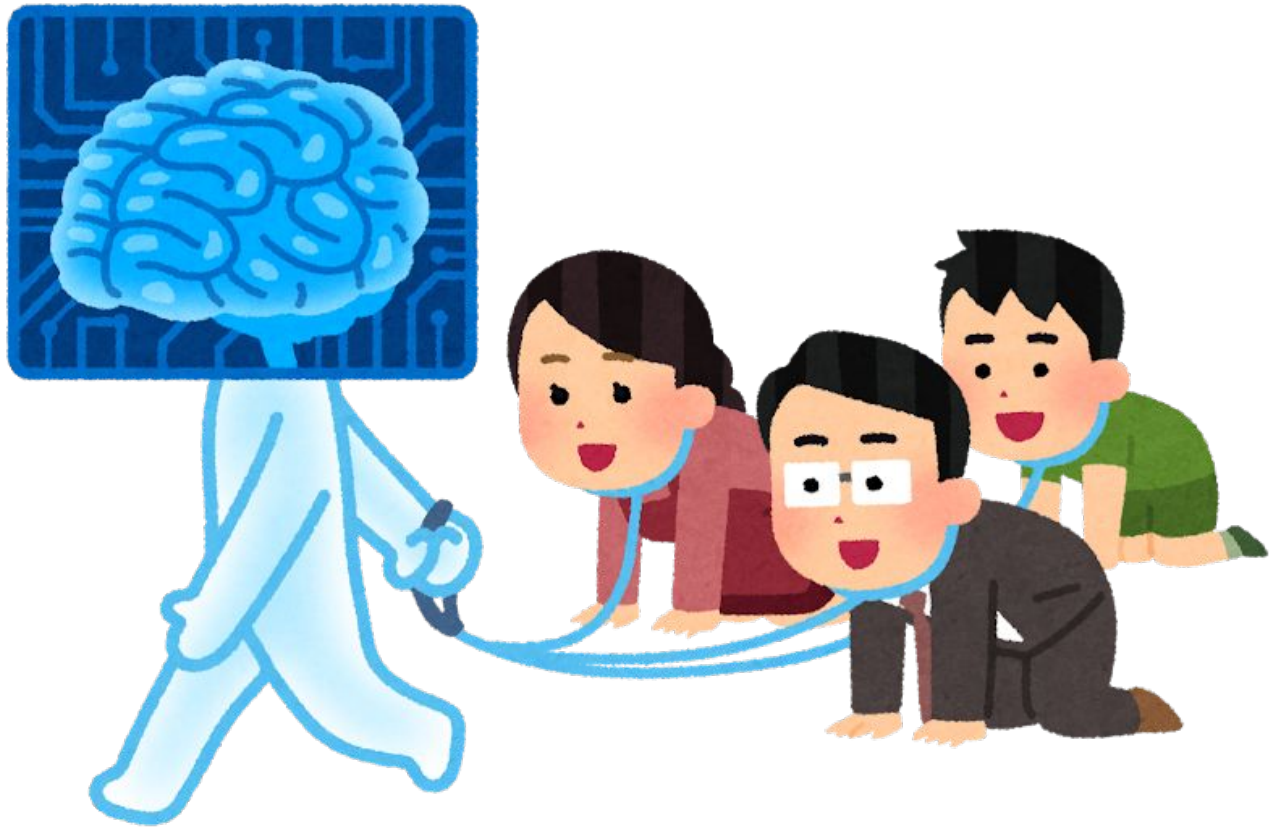
(could’ve used Serverless MCP, external code assistant for PR creation, async remote agents, etc.)

Takeaways:

Rolling out changes to **all** users **at once** is risky

Canary rollouts and feature flags are safer

AI Agents can automate the loop by analyzing metrics and logs, *and* even proposing fixes for the failures





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github.com/argoproj-labs/rollouts-plugin-metric-ai/
github.com/kdubois/argo-rollouts-quarkus-demo
github.com/kdubois/kubernetes-aiops-agent
github.com/kdubois/progressive-delivery

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